# Threat Assessment Report

Executive summary for model robustness under adversarial attack

## Model: 20260206_171419_resnet50_classifier.pt

Attack Type: DEEPFOOL
Images Evaluated: 10

| Attack Success Rate | Original Confidence | Adversarial Confidence | Execution Time |
|---|---|---|---|
| 100.0% | 53.43% | 30.01% | 22.83s |
| | | | Accuracy drop: 23.42% |

### Threat Level Assessment

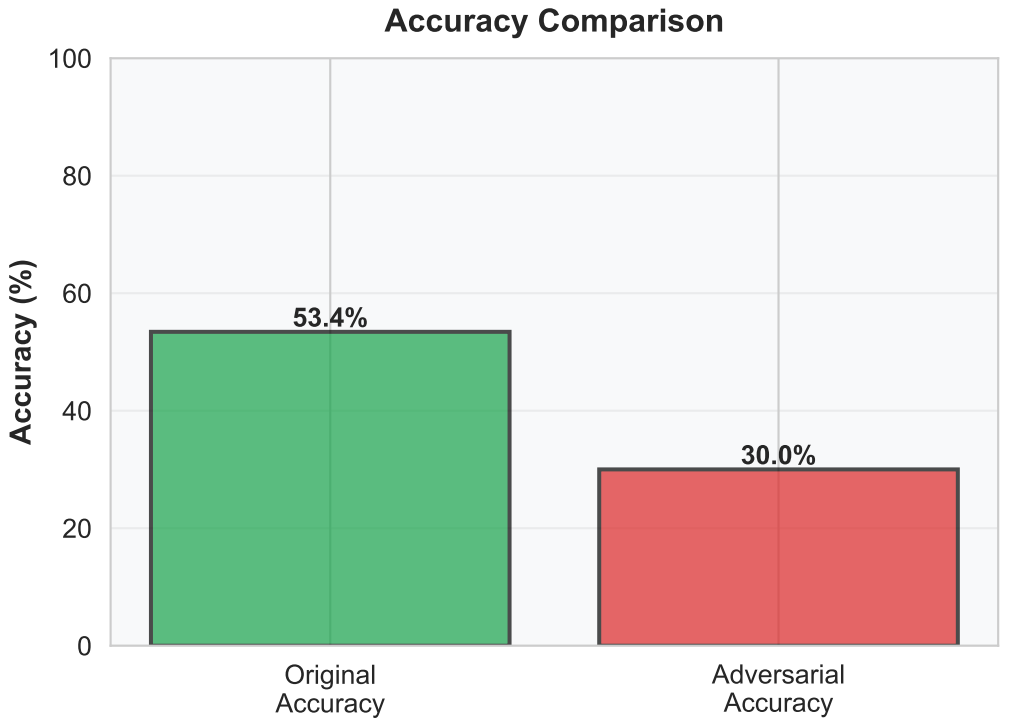
**HIGH RISK**

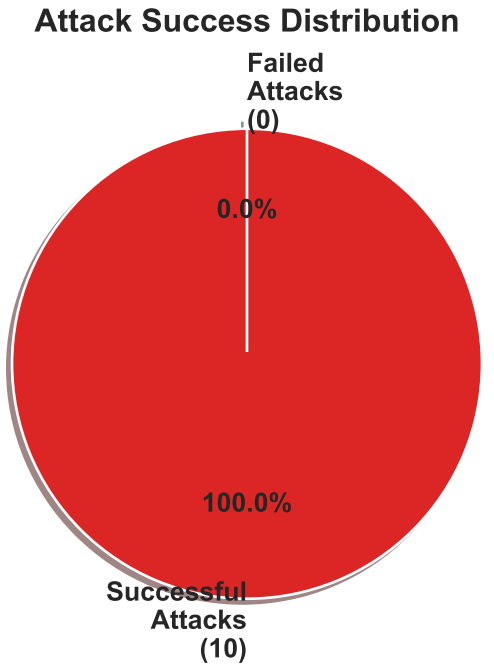
The DEEPFOOL attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 53.43% to 30.01%, indicating the current robustness posture is high risk.

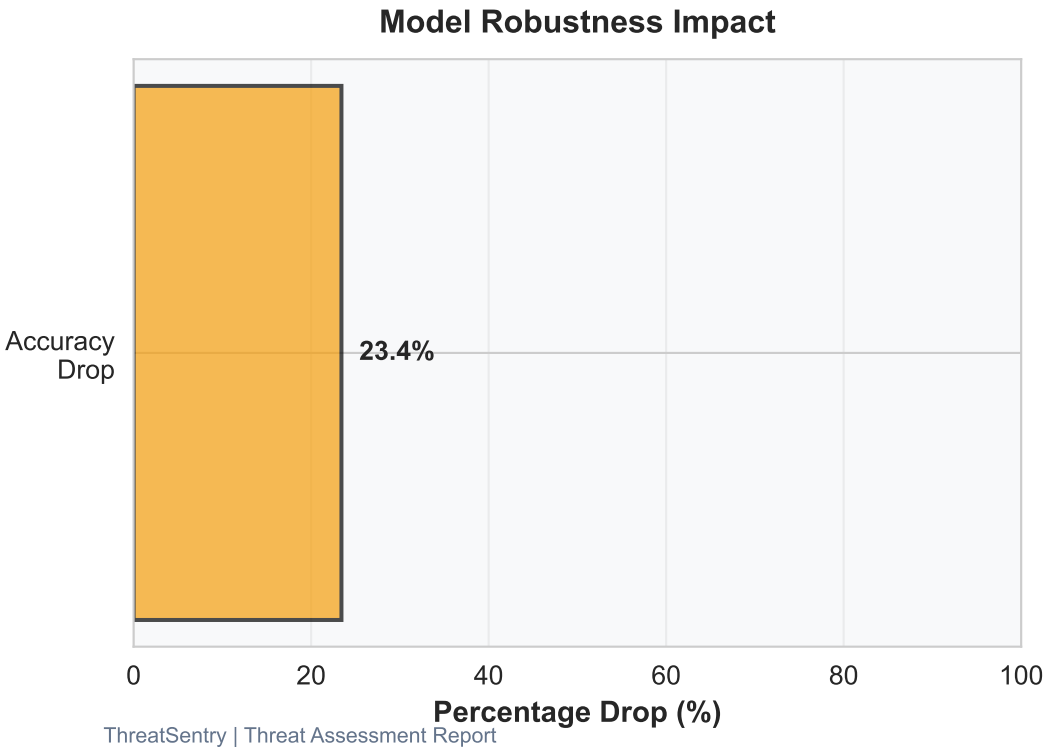
### Assessment Narrative

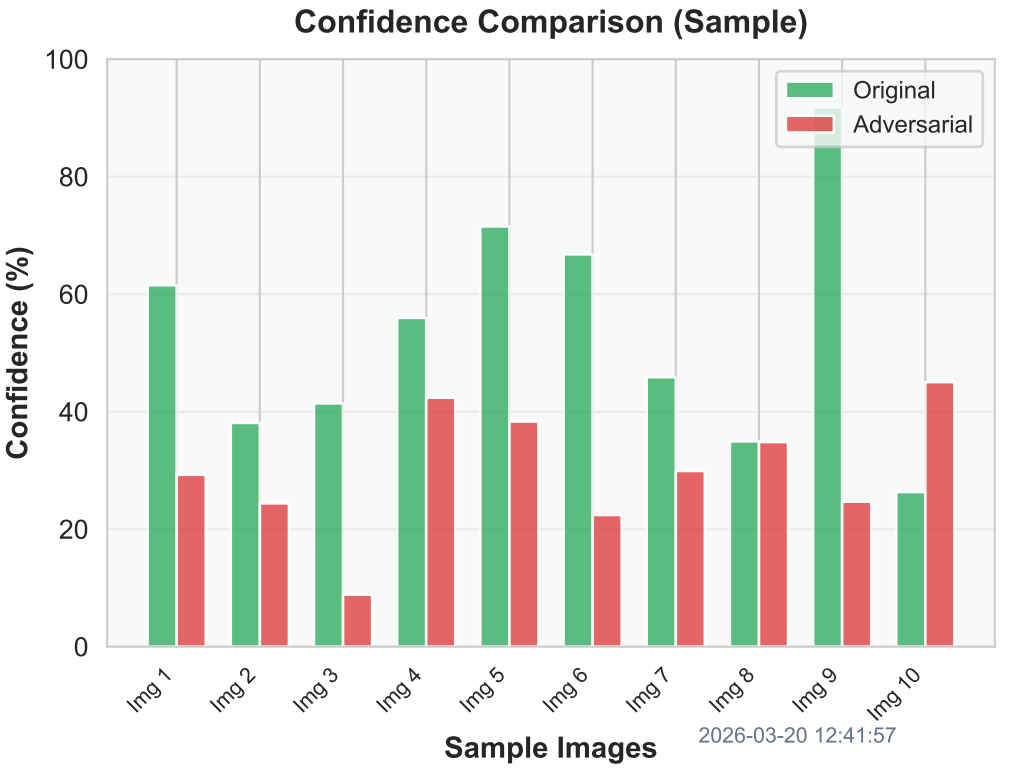
Successfully executed DEEPFOOL attack on model 20260206_171419_resnet50_classifier.pt using 10 test images. Attack success rate: 100.0%. Average original accuracy: 53.43%, Average adversarial accuracy: 30.01%. The attack successfully fooled the model in 10 out of 10 cases.

# Detailed Analysis



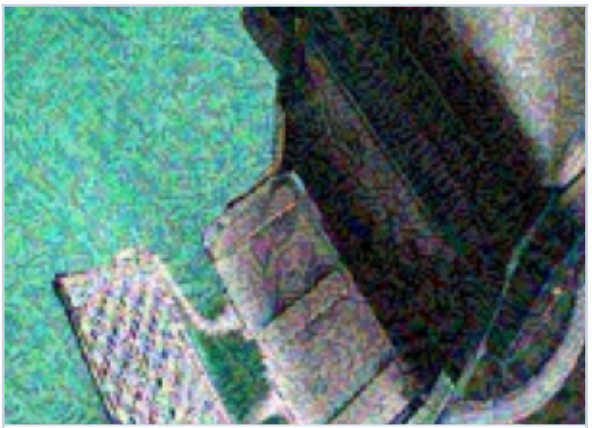



# Image Evidence and Predictions

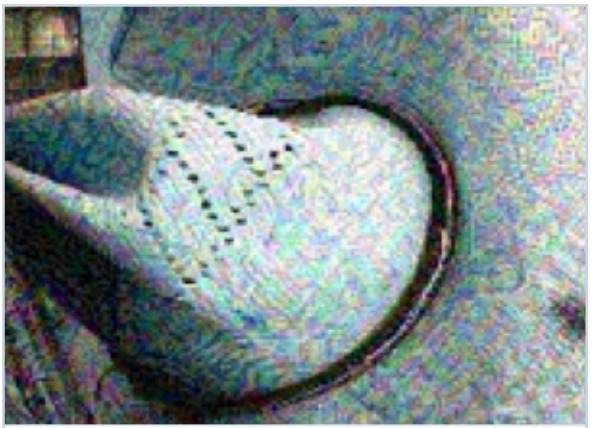
Images 1-4 of 10

**Attack Successful**
Original image: dugong.JPEG
Adversarial image: dugong.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 149
Original confidence: 91.81%
After attack predicted as: Class 109
Adversarial confidence: 24.68%

**Attack Successful**
Original image: tick.JPEG
Adversarial image: tick.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 76
Original confidence: 38.10%
After attack predicted as: Class 794
Adversarial confidence: 24.40%



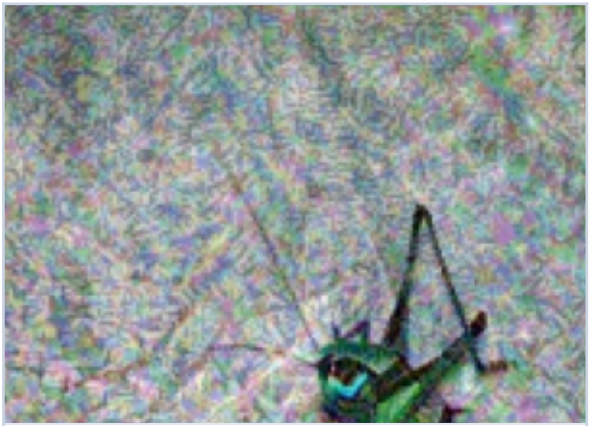
**Attack Successful**
Original image: barbershop.JPEG
Adversarial image: barbershop.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 597
Original confidence: 34.95%
After attack predicted as: Class 591
Adversarial confidence: 34.84%



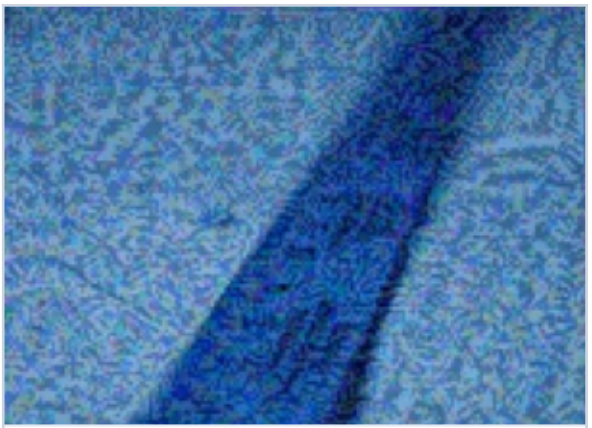
**Attack Successful**
Original image: sombrero.JPEG
Adversarial image: sombrero.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 515
Original confidence: 66.78%
After attack predicted as: Class 815
Adversarial confidence: 22.42%

# Image Evidence and Predictions

**Attack Successful**
Original image: tow_truck.JPEG
Adversarial image: tow_truck.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 627
Original confidence: 61.53%
After attack predicted as: Class 750
Adversarial confidence: 29.27%



**Attack Successful**
Original image: grasshopper.JPEG
Adversarial image: grasshopper.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 312
Original confidence: 41.42%
After attack predicted as: Class 411
Adversarial confidence: 8.85%

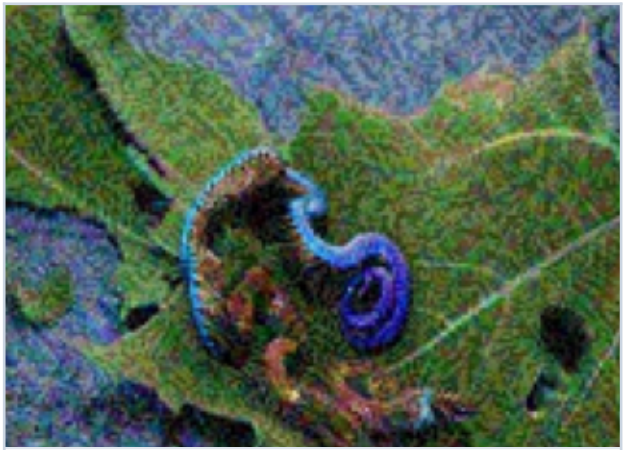


**Attack Successful**
Original image: rule.JPEG
Adversarial image: rule.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 911
Original confidence: 26.32%
After attack predicted as: Class 434
Adversarial confidence: 45.04%

**Attack Successful**
Original image: plow.JPEG
Adversarial image: plow.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 559
Original confidence: 45.88%
After attack predicted as: Class 750
Adversarial confidence: 29.92%

# Image Evidence and Predictions





**Attack Successful**
Original image: ringneck_snake.JPEG
Adversarial image: ringneck_snake.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 109
Original confidence: 71.54%
After attack predicted as: Class 549
Adversarial confidence: 38.32%

**Attack Successful**
Original image: mud_turtle.JPEG
Adversarial image: mud_turtle.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 973
Original confidence: 55.97%
After attack predicted as: Class 549
Adversarial confidence: 42.37%

# Recommendations

Security actions based on observed threat exposure

## Assessment Summary

Successfully executed DEEPFOOL attack on model 20260206_171419_resnet50_classifier.pt using 10 test images. Attack success rate: 100.0%. Average original accuracy: 53.43%, Average adversarial accuracy: 30.01%. The attack successfully fooled the model in 10 out of 10 cases.

## Security Recommendations

### 1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

### 2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

### 3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

### 4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

### 5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques